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EDITORIAL

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Designing the future of the fourth industrial revolution

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ABSTRACT

The Fourth Industrial Revolution (4IR) looms large on the horizon. It is characterised by the fusion of cyber, physical, and human realms, with the heart of it powered by artificial general intelligence (AGI). This revolution brings profound economic, societal, and ethical implications for humanity. Addressing its intricate challenges necessitates a shift toward higher-order and higher-dimensional empathy and creativity for humanity-centred design innovations. Traditional design thinking and methodologies, while once effective, now fall short amidst the complexities of the 4IR. Future designers must adeptly wield creative artificial intelligence for humanity-centred design and execute robust entrepreneurship and leadership to ensure a human-led design process towards the future.

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Navigating challenges to human values

Human civilisation's trajectory unfolds through four industrial revolutions—mechanization, electrification, digitalisation, and the dawn of 'intelligentization' (Figure 1). The advent of the fourth industrial revolution (4IR) marks a new phase propelled by the design of intelligent technologies. While artificial intelligence (AI) takes the limelight, the fourth industrial revolution encompasses more than just AI. Klaus Schwab, founder of the World Economic Forum, defines it as a 'technological revolution that blurs lines between physical, digital, and biological spheres' (Schwab 2017). This holistic perspective underscores the integration of cyber, physical, and human systems—an evolution brimming with both benefits and challenges.

As we design autonomous vehicles with life-and-death decisions, novel ethical dilemmas abound (Awad et al. 2018; Xue and Pang 2022). The fusion of human biology with technology, epitomised by human genome editing and brain implant chips, raises profound ethical questions and safety risks (Yu, Xue, and Barrangou 2021). Similarly, algorithms govern pivotal aspects of our lives, from social media content to professional services (Mittelstadt et al. 2016). Notably, the rise of generative AI and artificial general intelligence (AGI), exemplified by ChatGPT's virality, spotlights deep-seated concerns such as cybercrime,

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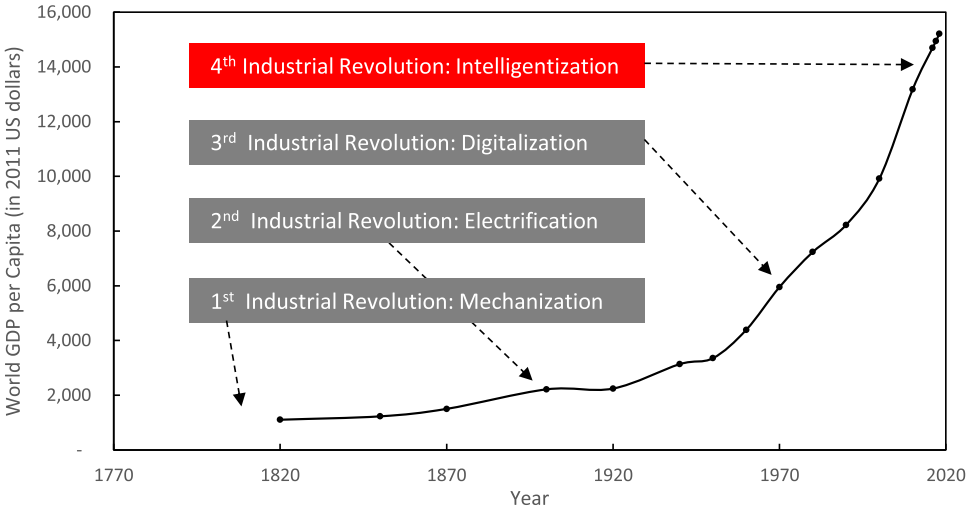


Figure 1. The four industrial revolutions. GDP data source: Maddison Project Database 2020 (Bolt and van Zanden 2020).

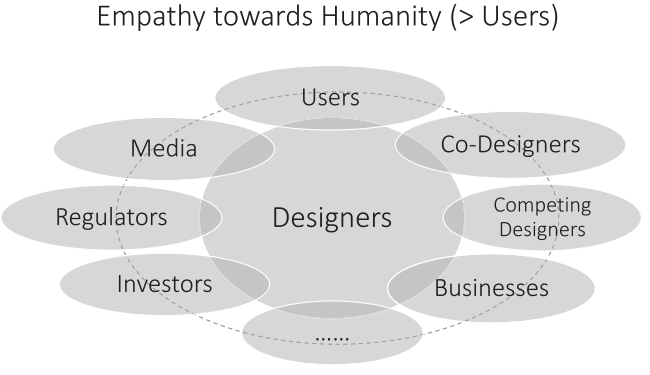


Figure 2. High-dimensional and high-order empathy towards diverse stakeholders.

misinformation, and job replacement (Altman 2023). This intricate landscape predominantly falls within the purview of industry titans like Google, Meta, and Microsoft, raising concerns of technology monopolies and global disparities.

In essence, the advent of 4IR introduces intricate challenges that interlace fundamental human values with far-reaching economic, societal, and ethical consequences. These issues are sociotechnical, involving a multitude of diverse stakeholders (Figure 2). Perhaps the most significant challenge is that innovation outpaces governance and reflection on potential issues in the 4IR. Traditional techniques for empathy development and user-centred design, such as surveys, focus groups and ethnographic observations, are inadequate to address these high-order and high-dimensional challenges to the collective values of humanity.

The design in the 4IR calls for an integrative human-cyber-physical system view from the outset, incorporating latent human values such as equality, justice, security, privacy, and ethics in the initial design requirements, and preparing for unexpected high-order emergent outcomes in these high-dimensional spaces while accounting for diverse stakeholders. Therefore, designers in the 4IR require empathy in higher dimensions and higher orders. However, even basic empathy remains an intuitive and often artistic skill for most designers (Brown 2008).

In light of the above, I make the following calls to action:

- Designers must respond and design solutions for the emergent challenges induced by the 4IR, facing humanity. These solutions will likely be sociotechnical and deeply systemic.
- Design needs to incorporate latent human values such as equality, justice, security, privacy, and ethics right from initial design requirements. These should be considered alongside explicit functionalities and costs. Traditional design models fall short in this aspect.
- Design demands high-order and high-dimensional empathy. However, empathy is fundamentally an intuitive and artistic skill.

Coping with complexity of system-of-systems and ecosystems

Design, as a human activity for deliberate transformation and value creation, has shaped our increasingly artificial world (Luo 2015a; Papalambros 2015). Now, our continuous design efforts have led to mounting complexity, necessitating a paradigm shift from reductionist methodologies to more holistic approaches to sustain innovation (Friedman et al. 2014). Traditional approaches of decomposing and optimising subsystems within existing systems fall short when grappling with the intricate interplay of emergent technologies. Increasingly, innovation emerges through the synthesis of disparate systems, transcending conventional design boundaries (Buchanan 2001).

In this context, complex sociotechnical systems or systems-of-systems emerge, bringing both unprecedented potential and multifaceted challenges (de Weck, Roos, and Magee 2016; Luo 2017; Norman and Stappers 2015). The design of 'intelligent connected electric vehicles' serve as an emblem of the paradox, aiming to enhance mobility, reduce emissions, and elevate quality of life, yet simultaneously raising pressing questions about costly infrastructure, cyber security, and socio-economic consequences.

Designing in this complex landscape requires a paradigm shift in design approaches and goals, where 'ilities' (Moses 2004) such as sustainability, reliability and evolvability seamlessly interweave with technical functionality and costs (Cutcher-Gershenfeld et al. 2004; Leveson et al. 2004; Luo 2015b). Coping with the complex and uncertain environments demands a brand of creativity that is high-dimensional and high-order, a quality notoriously elusive.

Here, I present a few additional calls to action:

- Design must adapt to the challenges presented by the emergence and complexity of system-of-systems. This adaptation could involve designing architectures, interfaces, standards, protocols, and more.

- Design should prioritise 'ilities' such as sustainability, reliability, and evolvability among other life-cycle properties. These should be incorporated right from the initial design requirements, alongside functionalities and costs. However, such 'ilities' can be challenging to quantify, measure and derive.
- Design of complex system-of-systems demands high-order and high-dimensional creativity. Yet, creativity inherently thrives on intuition and serendipity.

Embracing 'evolutionary ecosystem design thinking'

Design, as an agent of transformation, now operates within intricate ecosystems of interdependent, complementary, or competing co-designers (Baldwin 2022; Jacobides, Cennamo, and Gawer 2018; Luo et al. 2012). The evolution of systems of systems mandates that designers adopt an ecosystem-oriented mindset. Acknowledging the interdependence, symbiosis, competition, and cooperation among semi-autonomous co-design agents becomes paramount. Navigating this complex environment demands an evolutionary perspective – one that recognises unpredictability, fosters innovation from diverse quarters, adapts to ever-shifting conditions (Luo 2018).

This calls for an approach underpinned by empathy and creativity, which evolves into what can be termed *Evolutionary Ecosystem Design Thinking*. This approach demands high-order and high-dimensional capacities in both empathy and creativity to navigate the intricate web of complex and evolving interactions. Yet, nurturing these abilities proves to be a significant challenge for human designers.

Synergizing design and artificial general intelligence

Artificial intelligence, a hallmark of the 4IR, poses a paradigm shift for design and a pivotal force to address the complexities of the 4IR through design. While AI's development spans several decades, it is only recently that it has risen to the fore. Large Language Models (LLMs) such as the Generative Pretrained Transformers (GPT) showcase a remarkable fusion of cross-domain knowledge and versatile reasoning capabilities, akin to what can be called artificial generative intelligence (AGI).

Through both deep learning and deep generation, the AGI models exhibit capacities for empathy and creativity in extremely high dimensions and orders, sparking transformative possibilities in design (Verganti, Vendraminelli, and Iansiti 2020; Zhu and Luo 2023a). AGI's prowess has been evident in its capability to automate many tasks in the design process and solve intricate design challenges that outstrip human creativity (Makatura et al. 2023). For instance, with domain-specific fine-tuning and task-specific few-shot learning, GPT-based models can carry out *designerly thinking* to generate original and useful design concepts (Zhu and Luo 2023b; Zhu, Zhang, and Luo 2023).

However, the prospect of AI overshadowing human creative roles in the design process also beckons a cautionary note (Zhang et al. 2021). As AI's creative capabilities expand, it is conceivable that AI might come to dominate the design arena, marginalising human engagement. Preserving the relevance of human designers becomes a critical consideration. AI can indeed inspire, inform, and augment human creativity, but

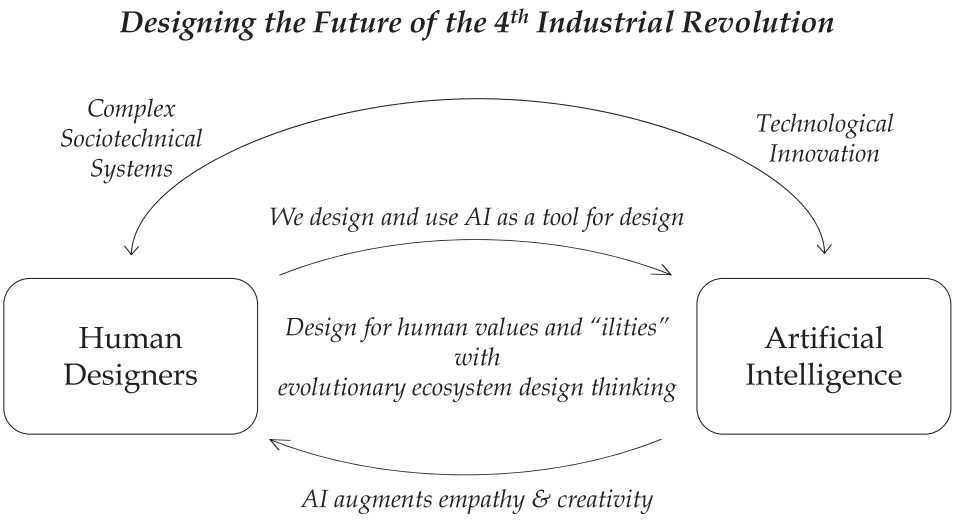


Figure 3. Designing the future of the 4th industrial revolution

retaining humans' roles as initiators and leaders of the design process remains pivotal (Luo 2023).

Forging a harmonious future

In summary, designing the future of the 4IR demands:

- Design that encompasses ethical and societal values and addresses 'ilities' for all stakeholders.
- Adoption of 'evolutionary ecosystem design thinking'.
- Leveraging AI to elevate dimensions and orders of empathy and creativity in design.
- Ensuring AI serves as a design tool, while upholding human leadership in the design process.

As AI's potential converges with the intricates of the 4IR, designers must wield AI as an augmentative tool rather than relinquishing control (Figure 3). The fusion of creative AI with human ingenuity holds the power to propel humanity-centred design, safeguard core human values, and tackle multifaceted challenges through high-order and high-dimensional empathy and creativity. However, preserving the essence of human-led design is imperative. This mandates that AI remains as a regulated tool, maintaining its role as an enabler rather than an equal collaborator or substitute of human designers. Therefore, future designers must embrace robust entrepreneurial and leadership mindsets, nurturing a tomorrow where human intellect leads AI to shape the contours of the 4IR—a testament to a future centred around humanity's progress (Norman 2023).

Disclosure statement

No potential conflict of interest was reported by the author(s).

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